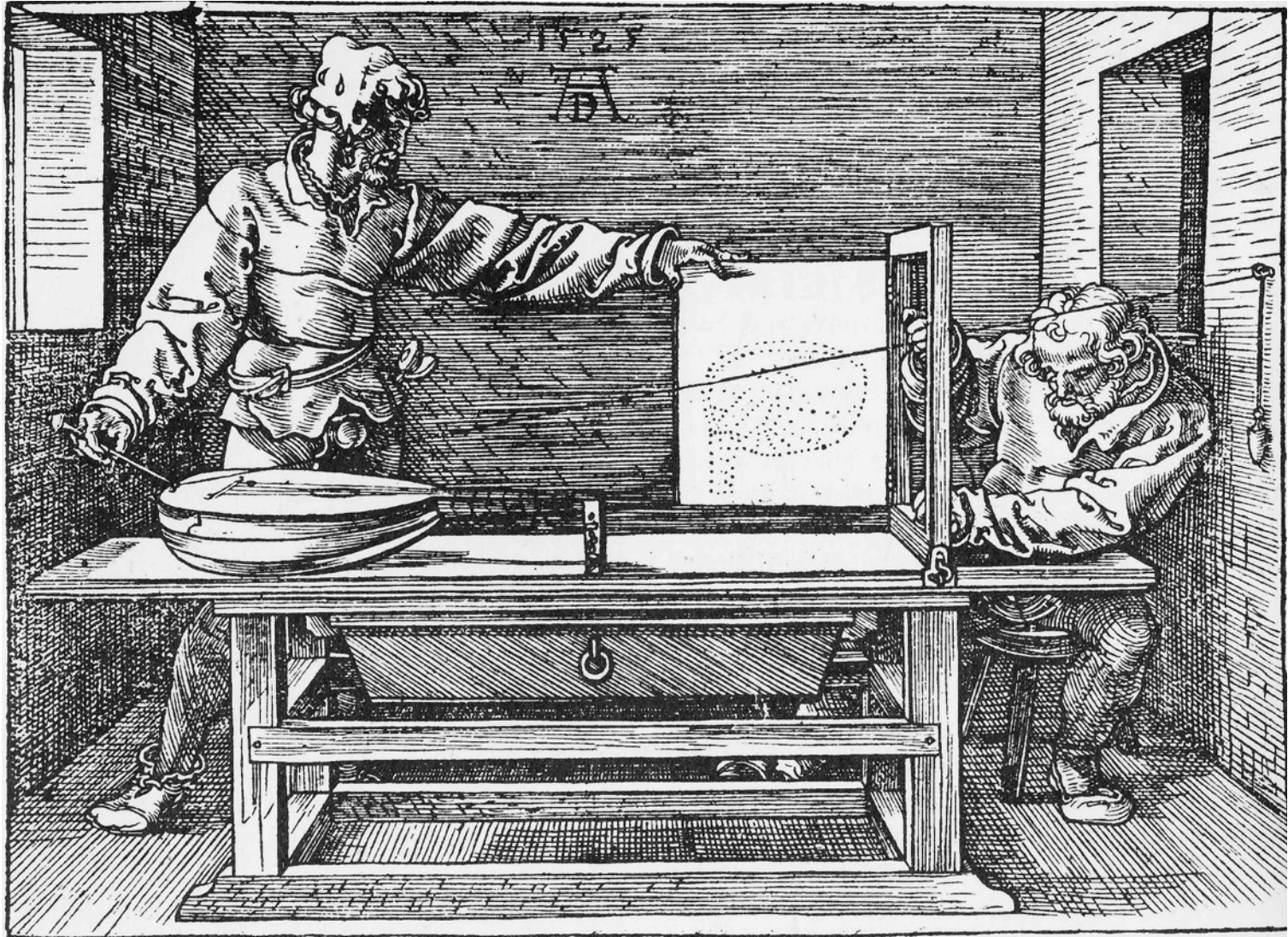


Perspective projection

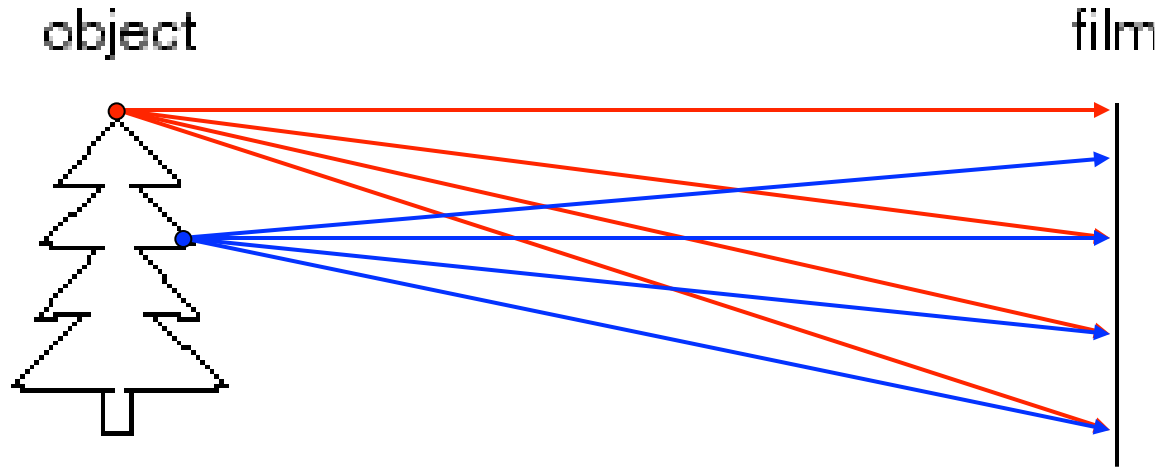


Albrecht Dürer, *Mechanical creation of a perspective image*, 1525

Overview of next two lectures

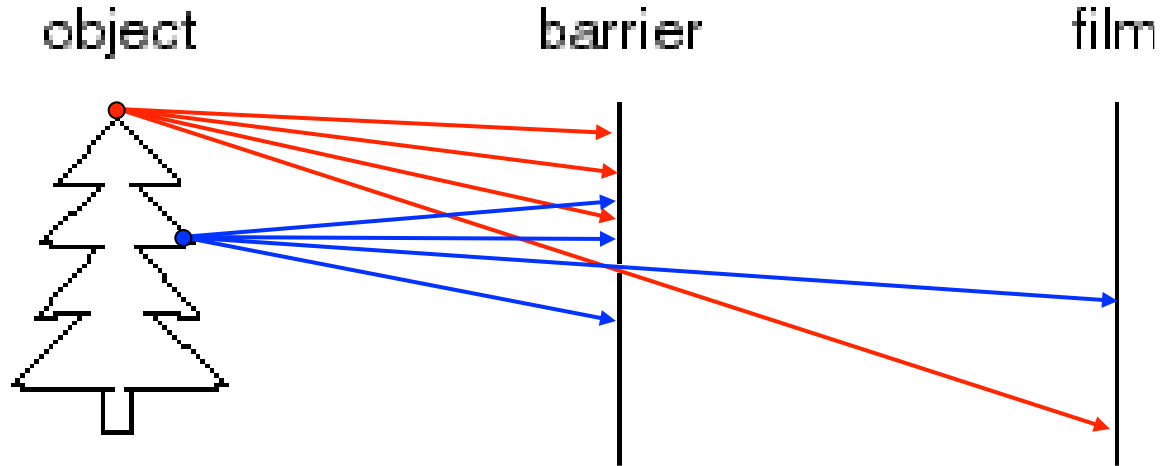
- The pinhole projection model
 - Qualitative properties
 - Perspective projection matrix
- Cameras with lenses
 - Depth of focus
 - Field of view
 - Lens aberrations
- Digital cameras
 - Sensors
 - Color
 - Artifacts

Let's design a camera



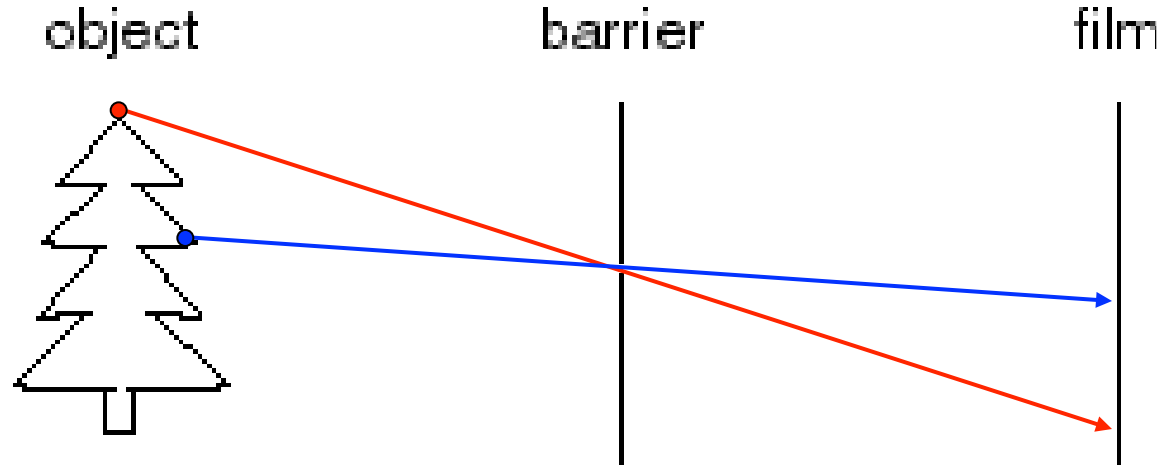
Idea 1: put a piece of film in front of an object
Do we get a reasonable image?

Pinhole camera



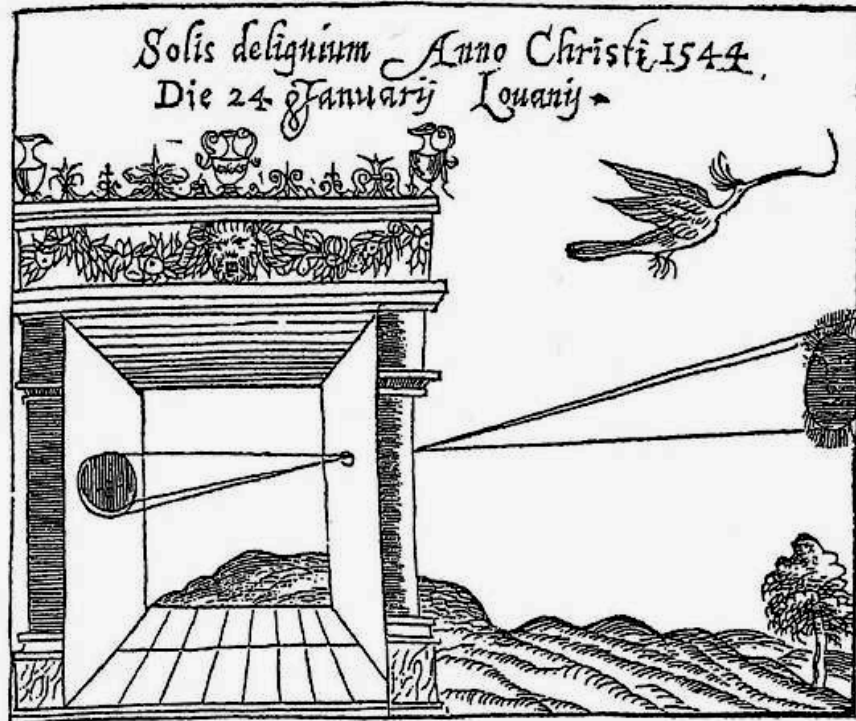
Add a barrier to block off most of the rays

Pinhole camera



- Captures **pencil of rays** – all rays through a single point: **aperture, center of projection, optical center, focal point, camera center**
- The image is formed on the **image plane**

Camera obscura



Gemma Frisius, 1558

- Basic principle known to Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)

Turning a room into a camera obscura



After scouting rooms and reserving one for at least a day, Morell masks the windows except for the aperture. He controls three elements: the size of the hole, with a smaller one yielding a sharper but dimmer image; the length of the exposure, usually eight hours; and the distance from the hole to the surface on which the outside image falls and which he will photograph. He used 4 x 5 and 8 x 10 view cameras and lenses ranging from 75 to 150 mm.

After he's done inside, it gets harder. "I leave the room and I am constantly checking the weather, I'm hoping the maid reads my note not to come in, I'm worrying that the sun will hit the plastic masking and it will fall down, or that I didn't trigger the lens."

From *Grand Images Through a Tiny Opening*, **Photo District News**, February 2005

Abelardo Morell, Camera Obscura Image of Manhattan View Looking South in Large Room, 1996

http://www.abelardomorell.net/camera_obscura1.html

Turning a room into a camera obscura

My hotel room,
contrast enhanced.



The view from my window

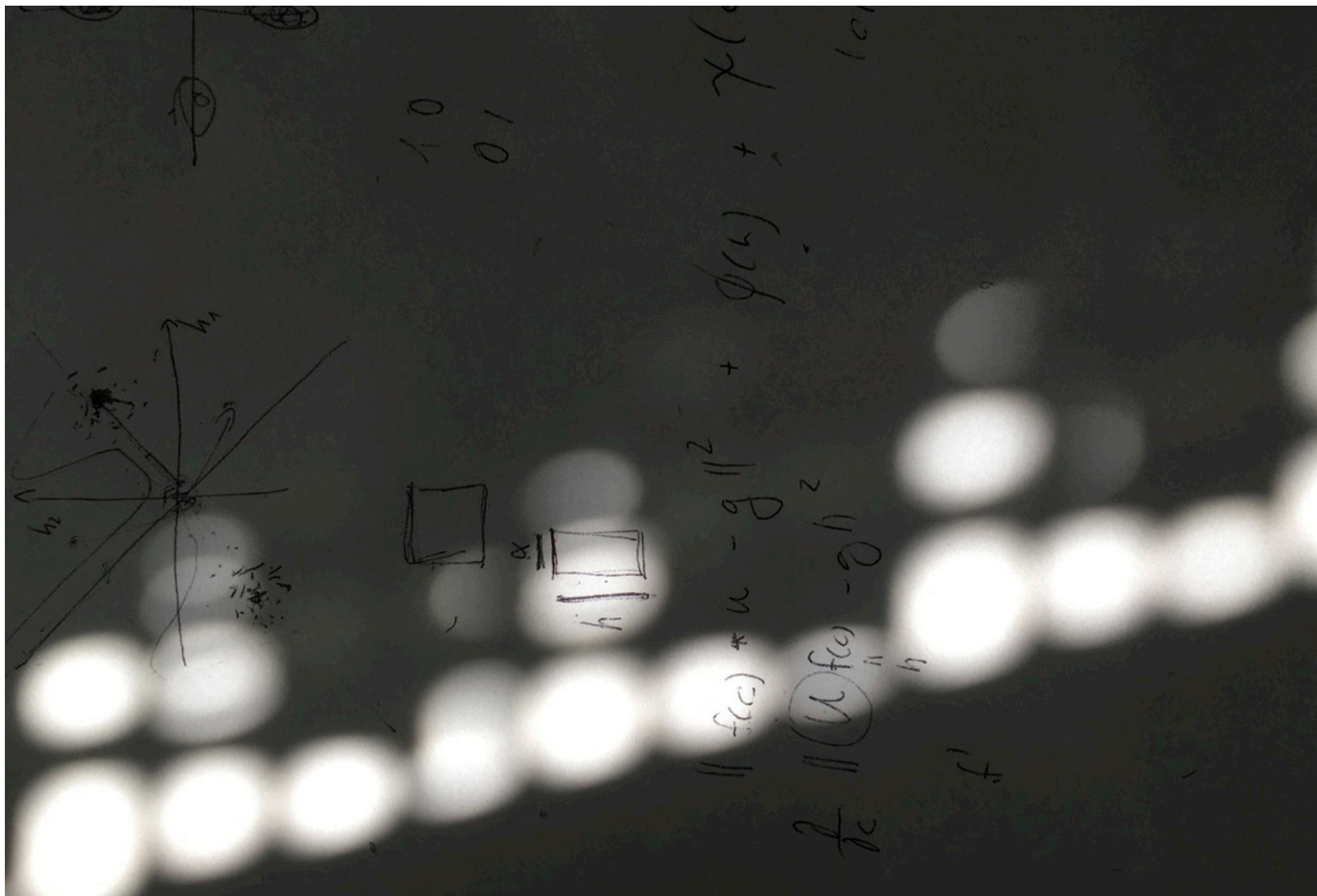


Accidental pinholes produce images that are
unnoticed or misinterpreted as shadows

Pinhole cameras everywhere



Pinhole cameras everywhere



Pinhole cameras everywhere



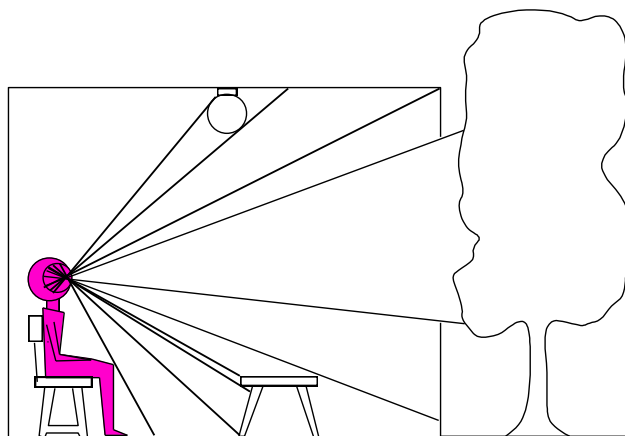
Tree shadow during a solar eclipse

photo credit: Nils van der Burg

<http://www.physicstogo.org/index.cfm>

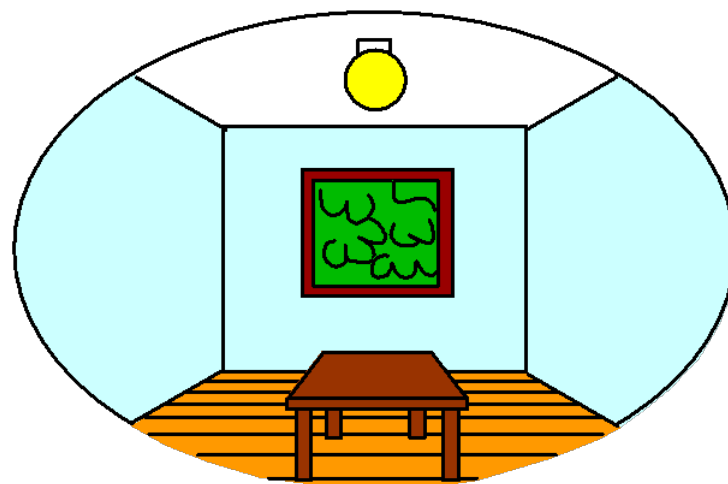
Dimensionality reduction: from 3D to 2D

3D world



Point of observation

2D image



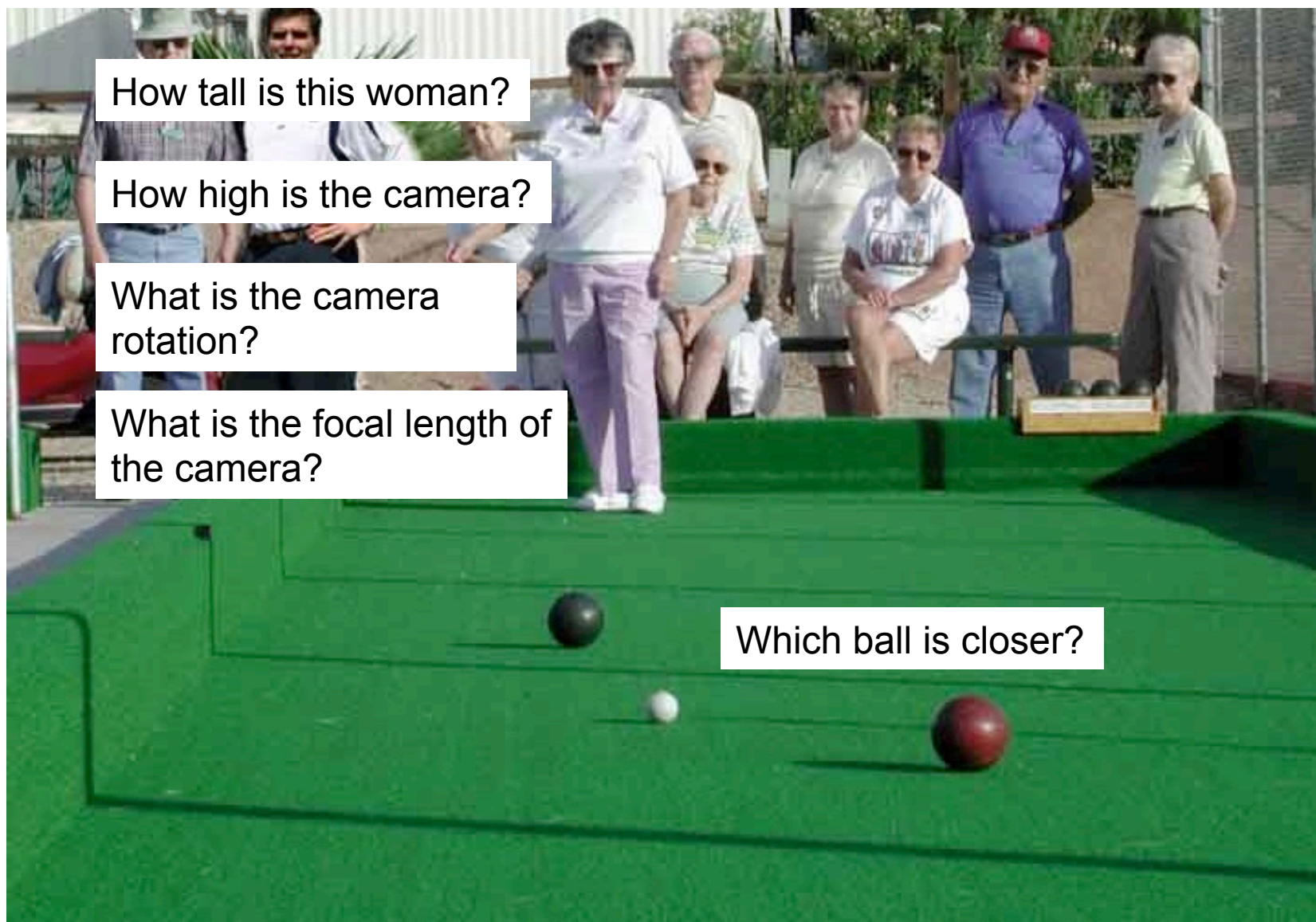
What properties of the world are preserved?

- Straight lines, incidence

What properties are not preserved?

- Angles, lengths

Single-view Geometry



How tall is this woman?

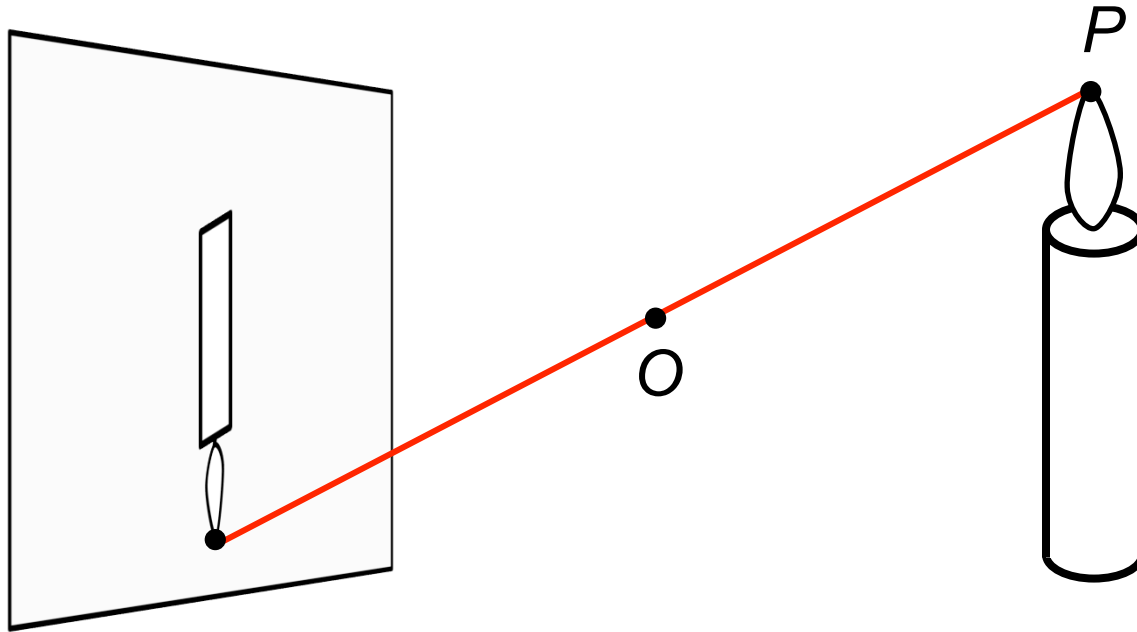
How high is the camera?

What is the camera rotation?

What is the focal length of the camera?

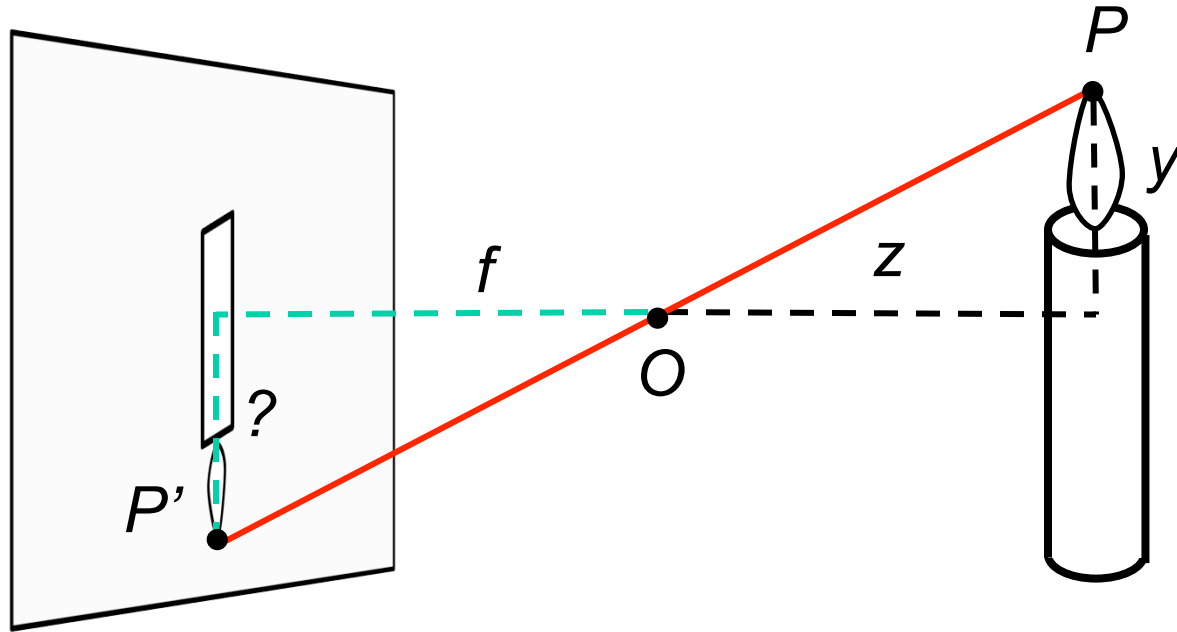
Which ball is closer?

Modeling projection



- To compute the projection P' of a scene point P , form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
 - All scene points that lie on this visual ray have the same projection in the image

Modeling projection



The coordinate system

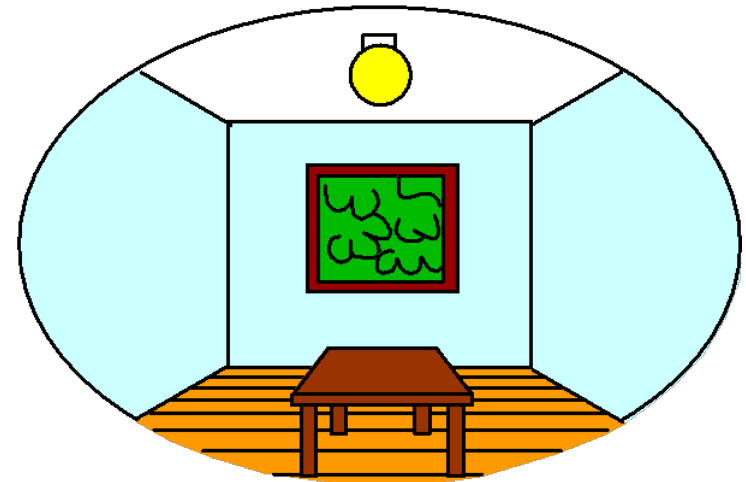
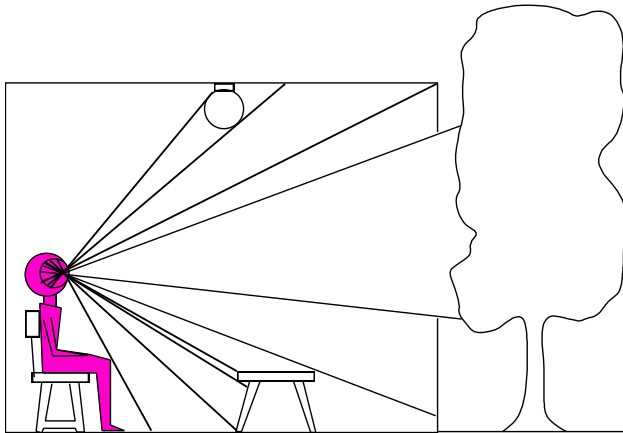
- The optical center (O) is at the origin
- The image plane is parallel to xy -plane or perpendicular to the z -axis, which is the *optical axis*

Projection equations

- Derived using similar triangles $(x, y, z) \rightarrow (f \frac{x}{z}, f \frac{y}{z})$

Fronto-parallel planes

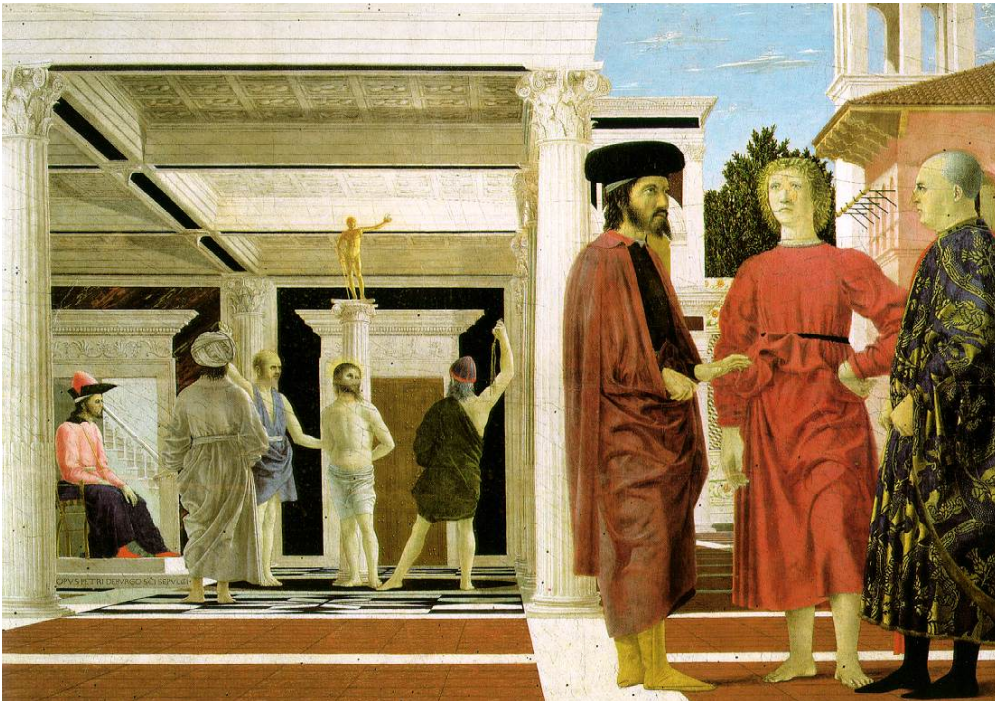
- What happens to the projection of a pattern on a plane parallel to the image plane?
 - All points on that plane are at a fixed *depth* z
 - The pattern gets scaled by a factor of f / z , but angles and ratios of lengths/areas are preserved



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

Fronto-parallel planes

- What happens to the projection of a pattern on a plane parallel to the image plane?
 - All points on that plane are at a fixed *depth* z
 - The pattern gets scaled by a factor of f / z , but angles and ratios of lengths/areas are preserved

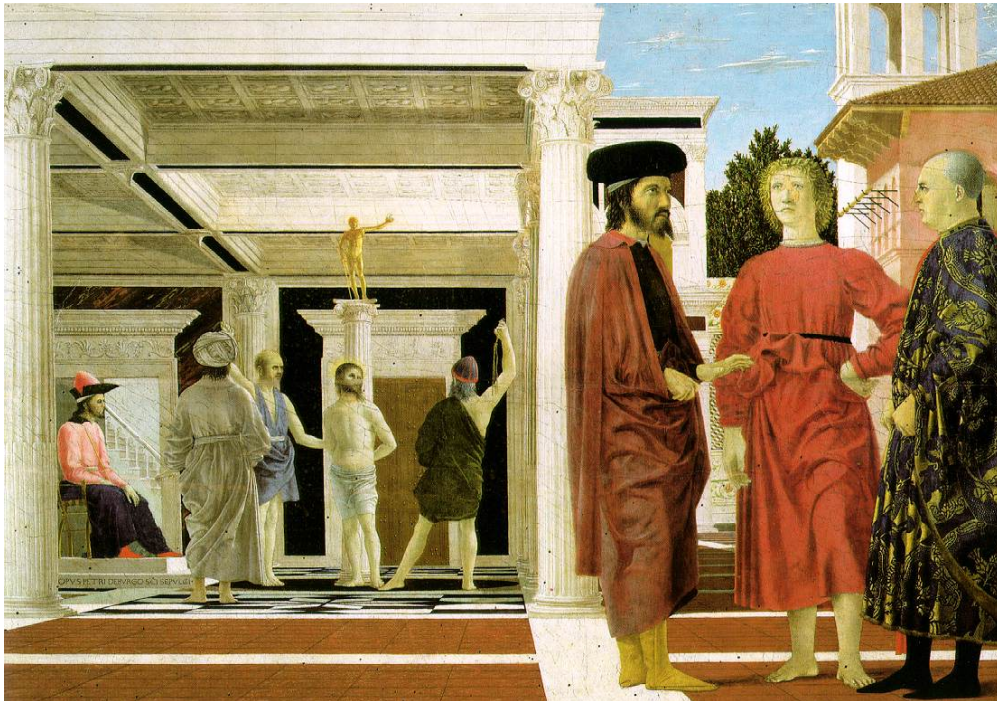


Piero della Francesca, *Flagellation of Christ*, 1455-1460



Jan Vermeer, *The Music Lesson*, 1662-1665

What about non-fronto-parallel planes?



Piero della Francesca, *Flagellation of Christ*, 1455-1460



Jan Vermeer, *The Music Lesson*, 1662-1665

Projection can be tricky...



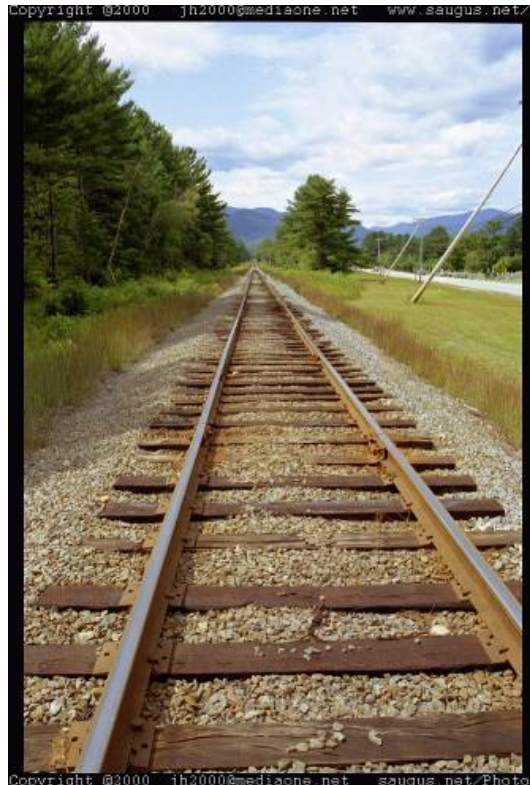
Projection can be tricky...



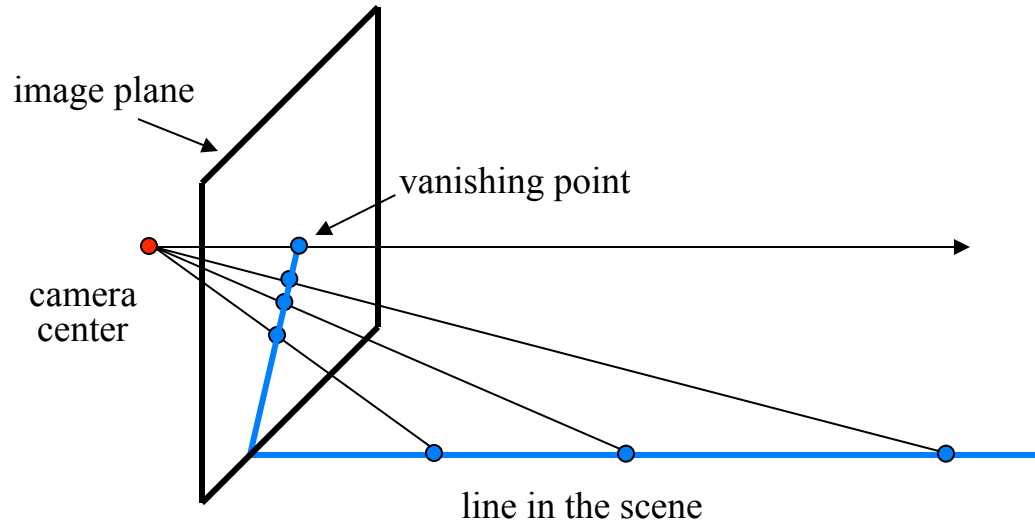
Making of 3D sidewalk art:

Vanishing points

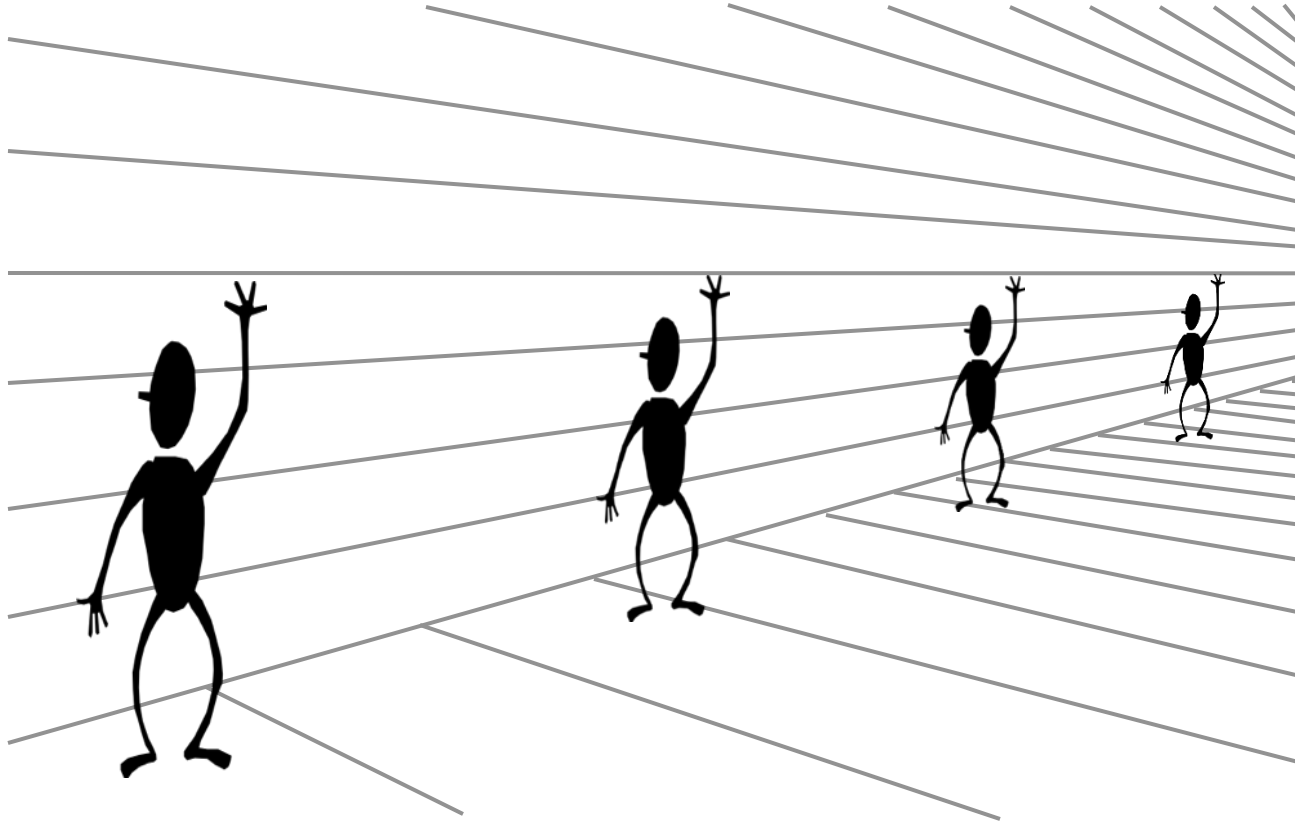
- All parallel lines converge to a *vanishing point*
 - Each direction in space is associated with its own vanishing point
 - Exception: directions parallel to the image plane



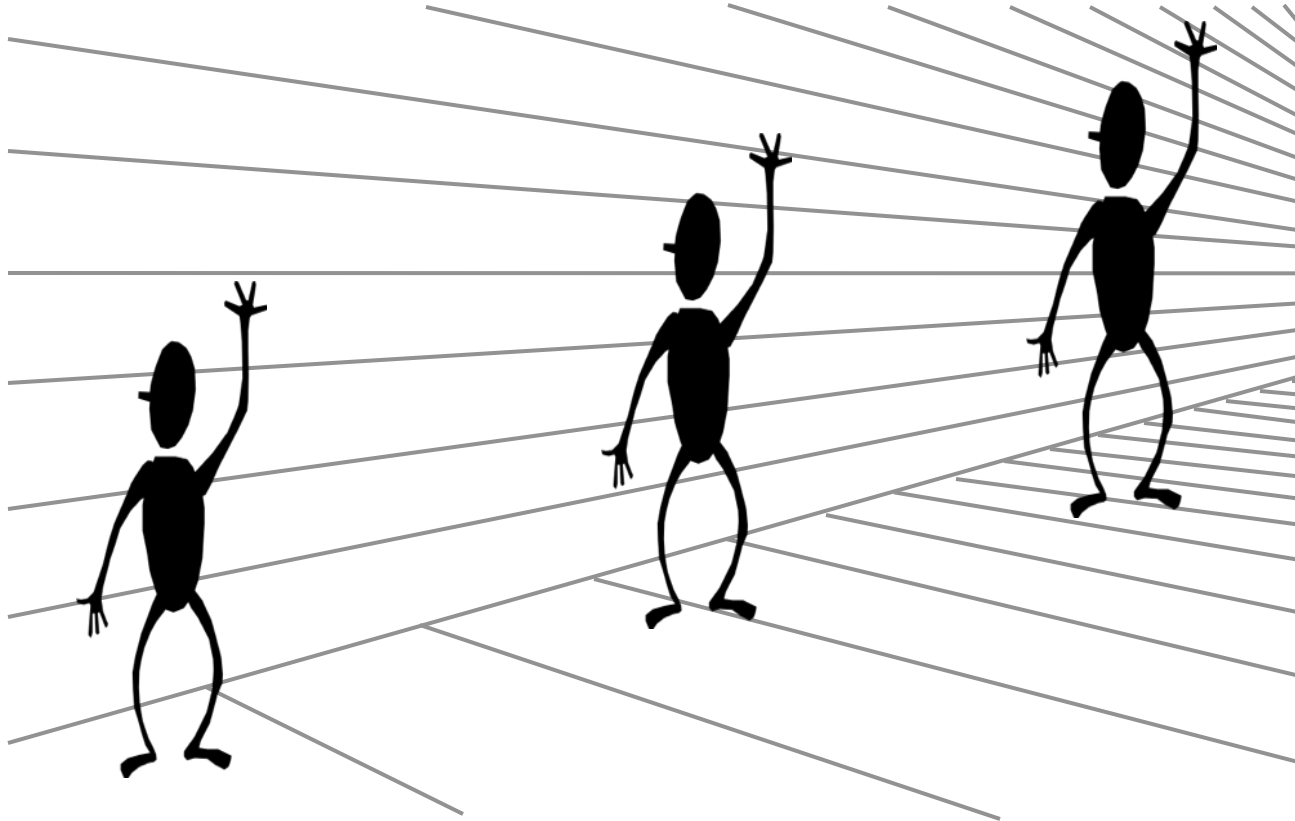
Constructing the vanishing point of a line



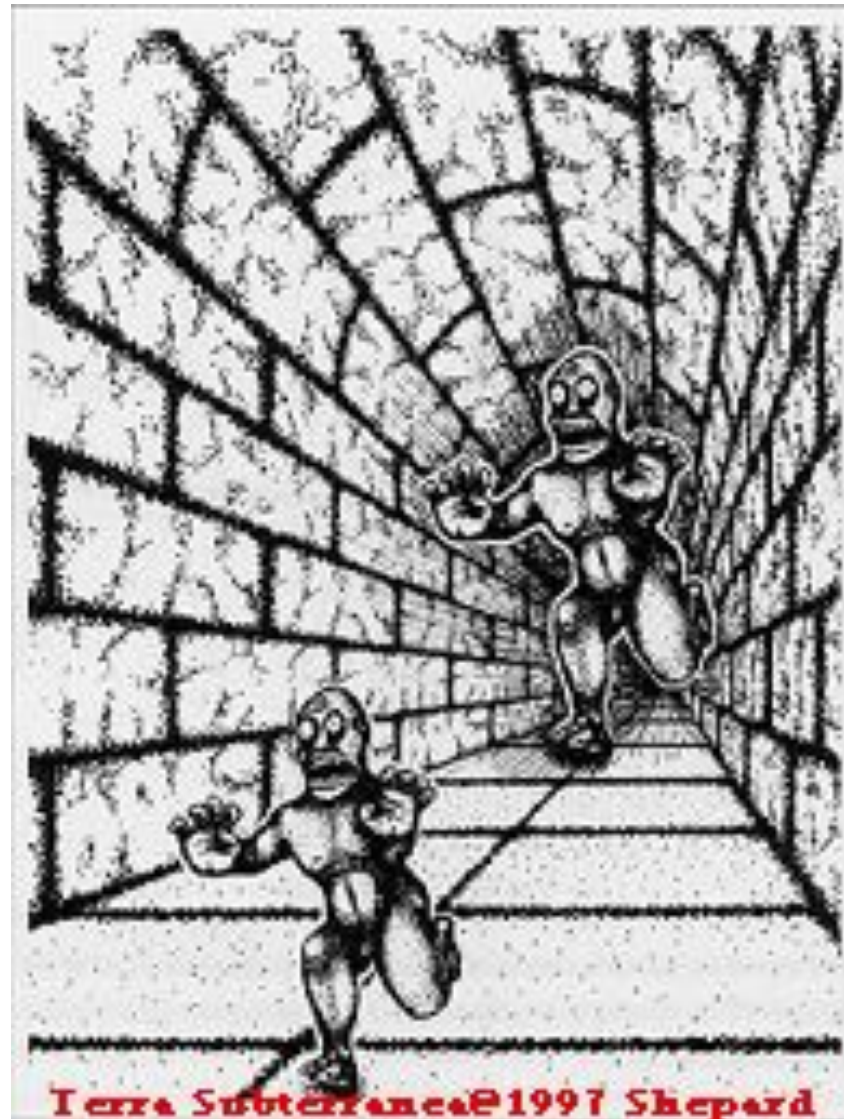
Perspective cues



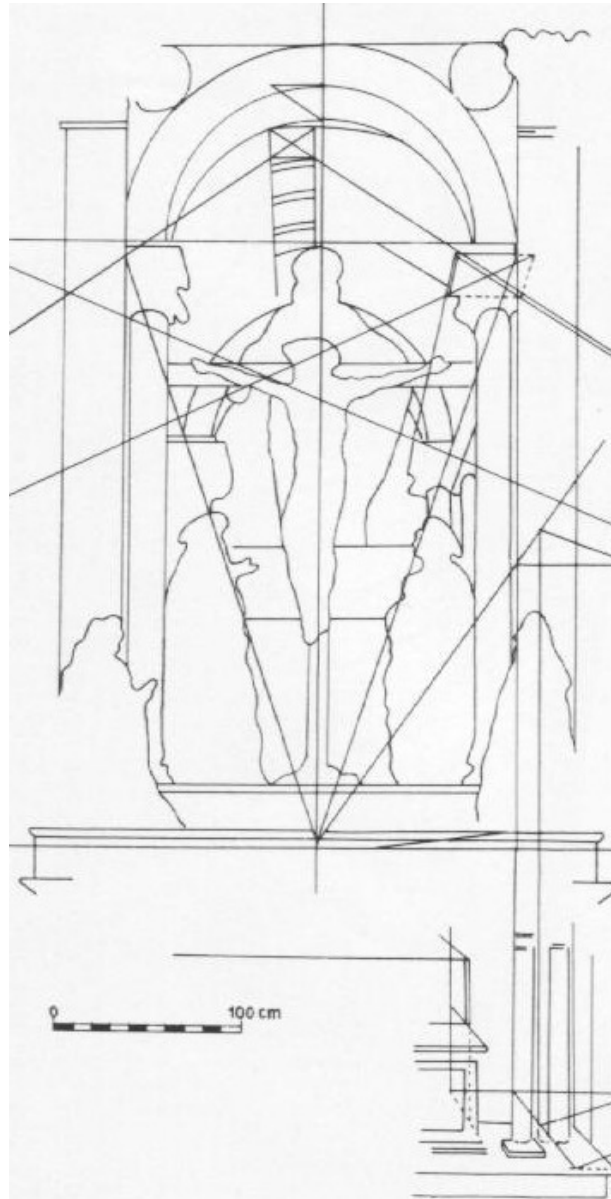
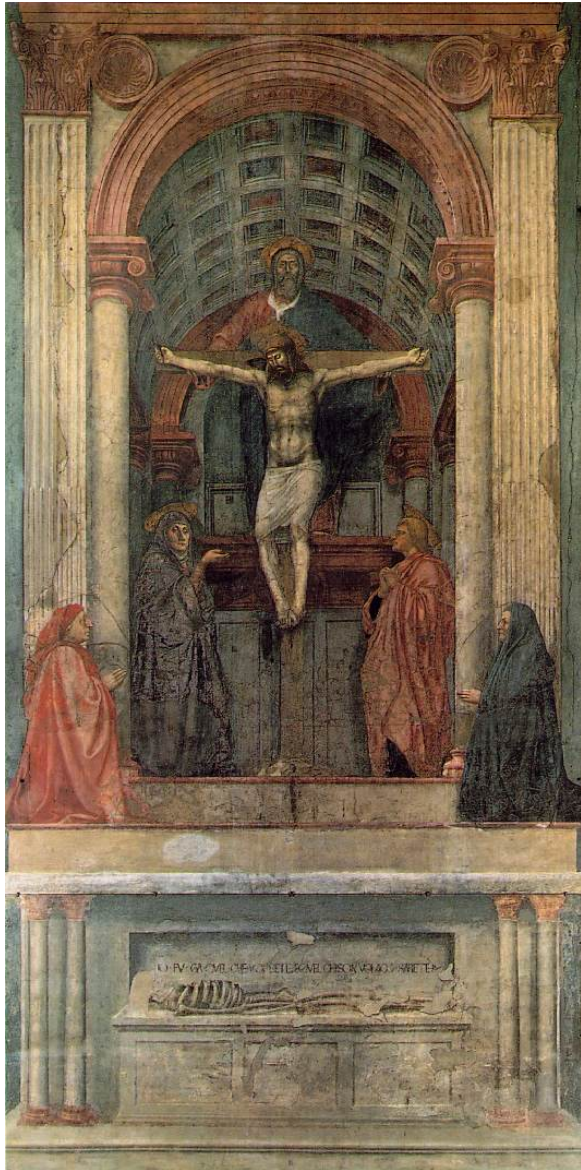
Perspective cues



Perspective cues



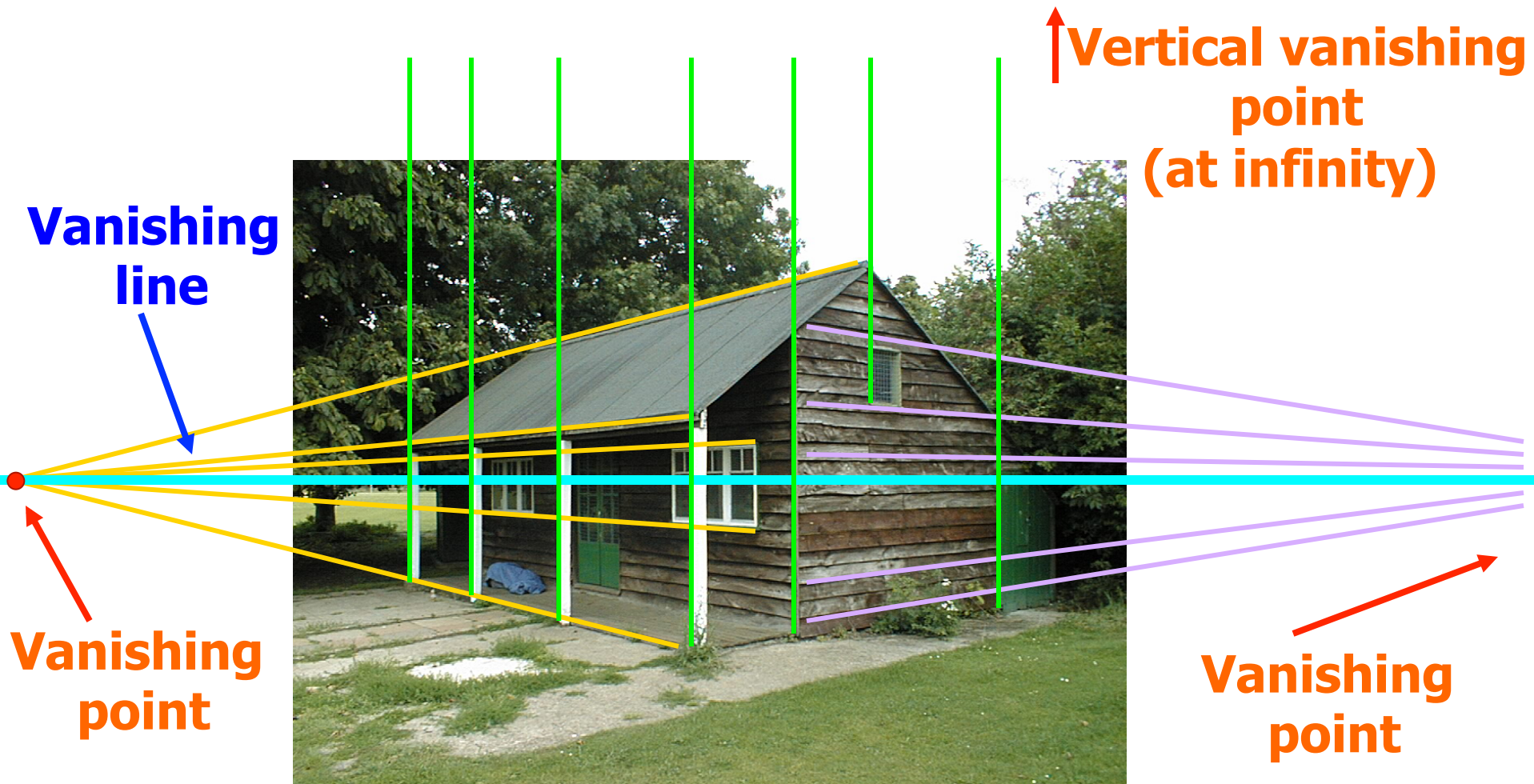
Perspective cues in art



Masaccio, *Trinity*, Santa Maria Novella, Florence, 1425-28

One of the first consistent uses of perspective in Western art

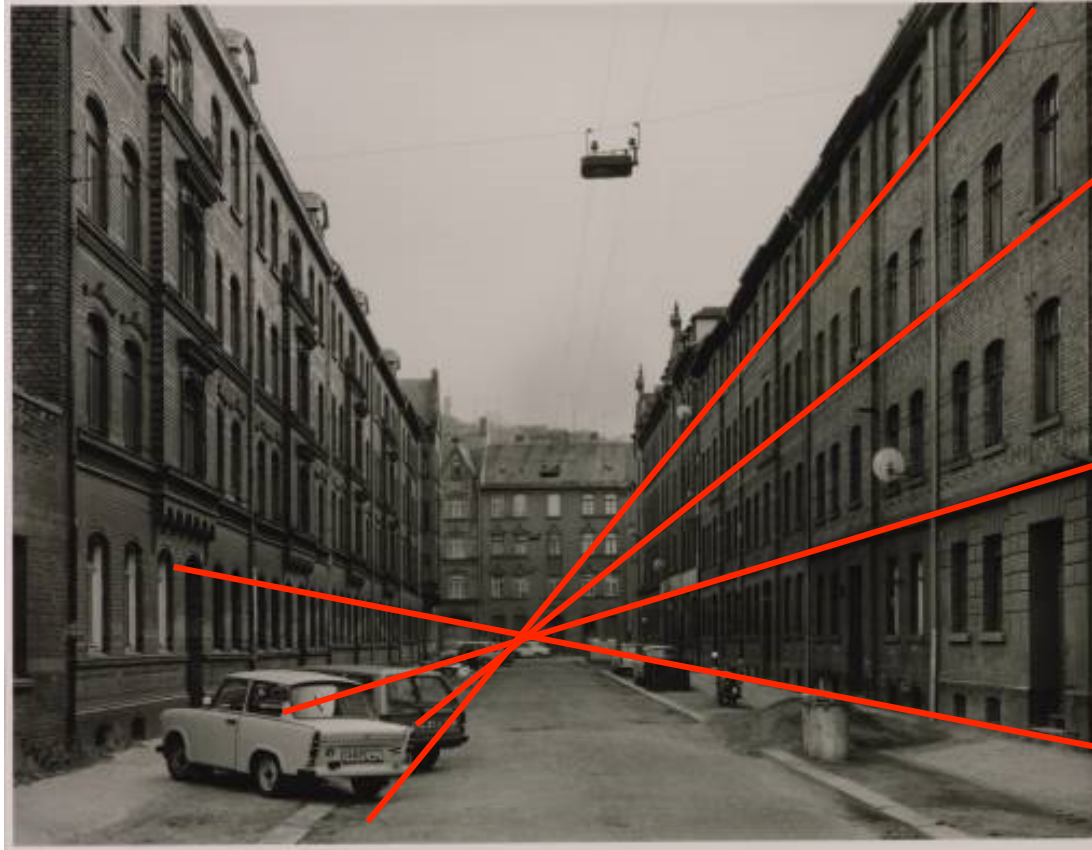
Vanishing points and lines



Vanishing points and lines



Note on estimating vanishing points



Use multiple lines for better accuracy

... but lines will not intersect at exactly the same point in practice

One solution: take mean of intersecting pairs

... bad idea!

Instead, minimize angular differences

Vanishing objects

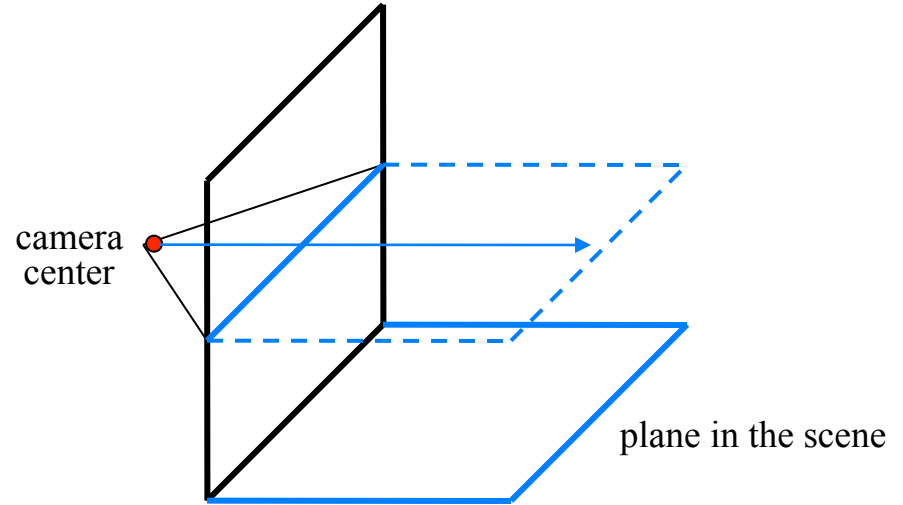
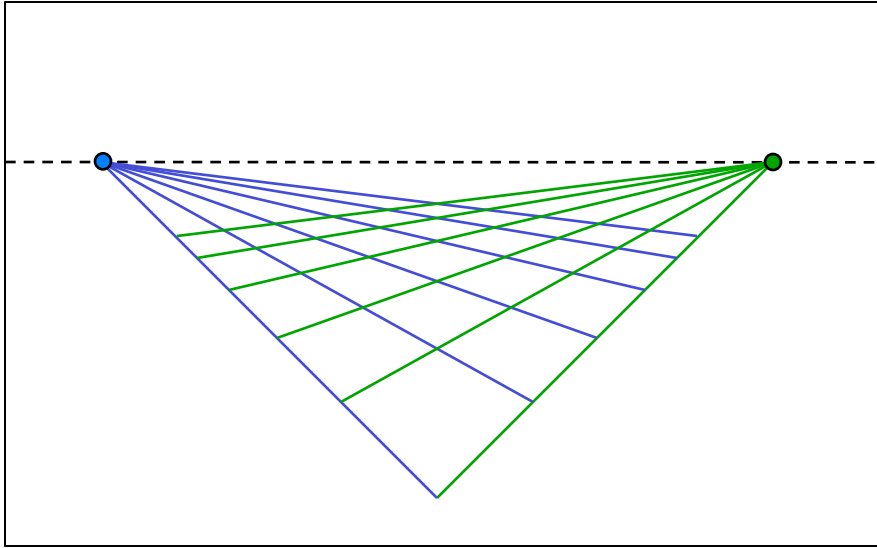


Vanishing lines of planes



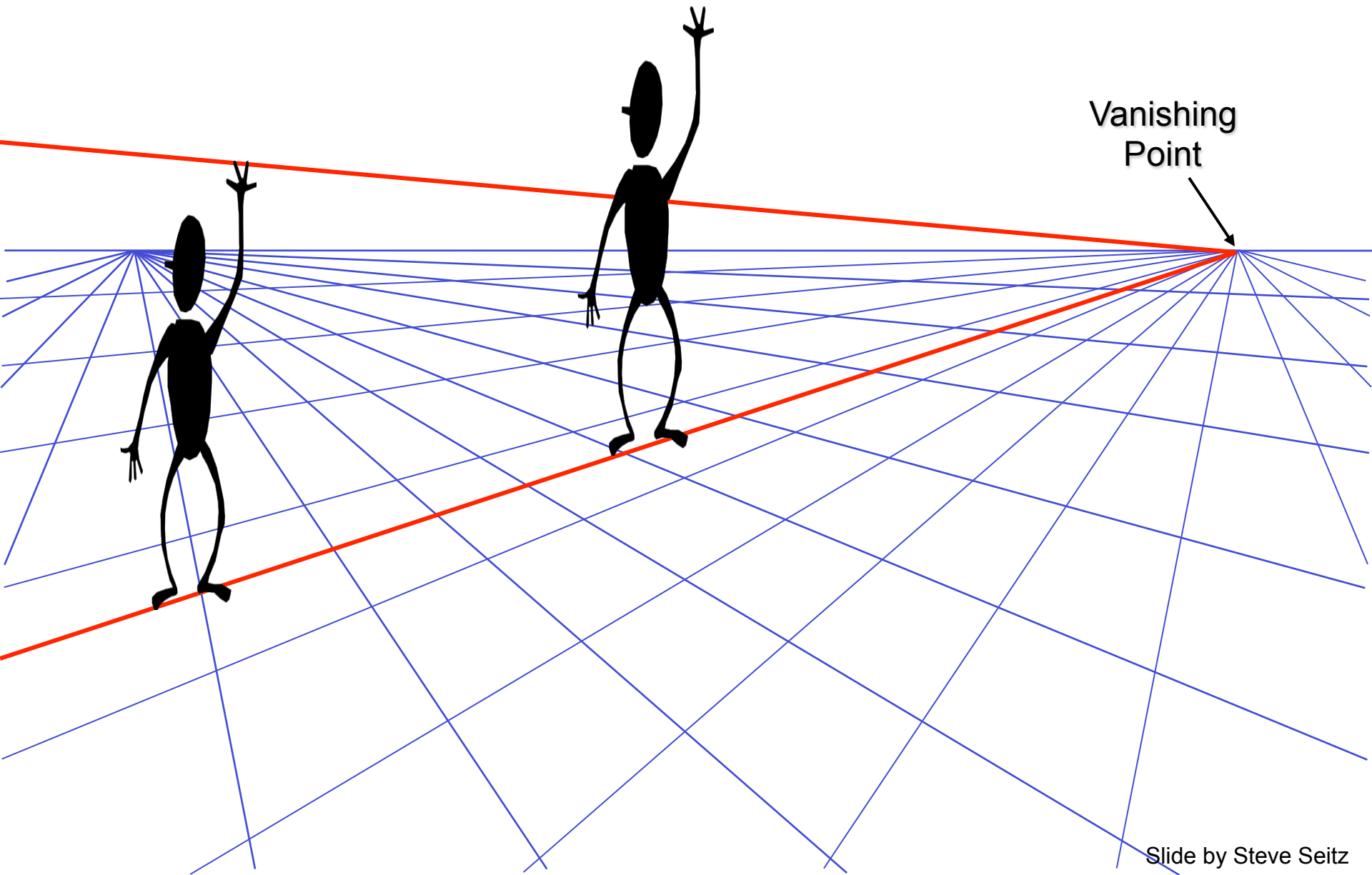
How do we construct the vanishing line of a plane?

Vanishing lines of planes

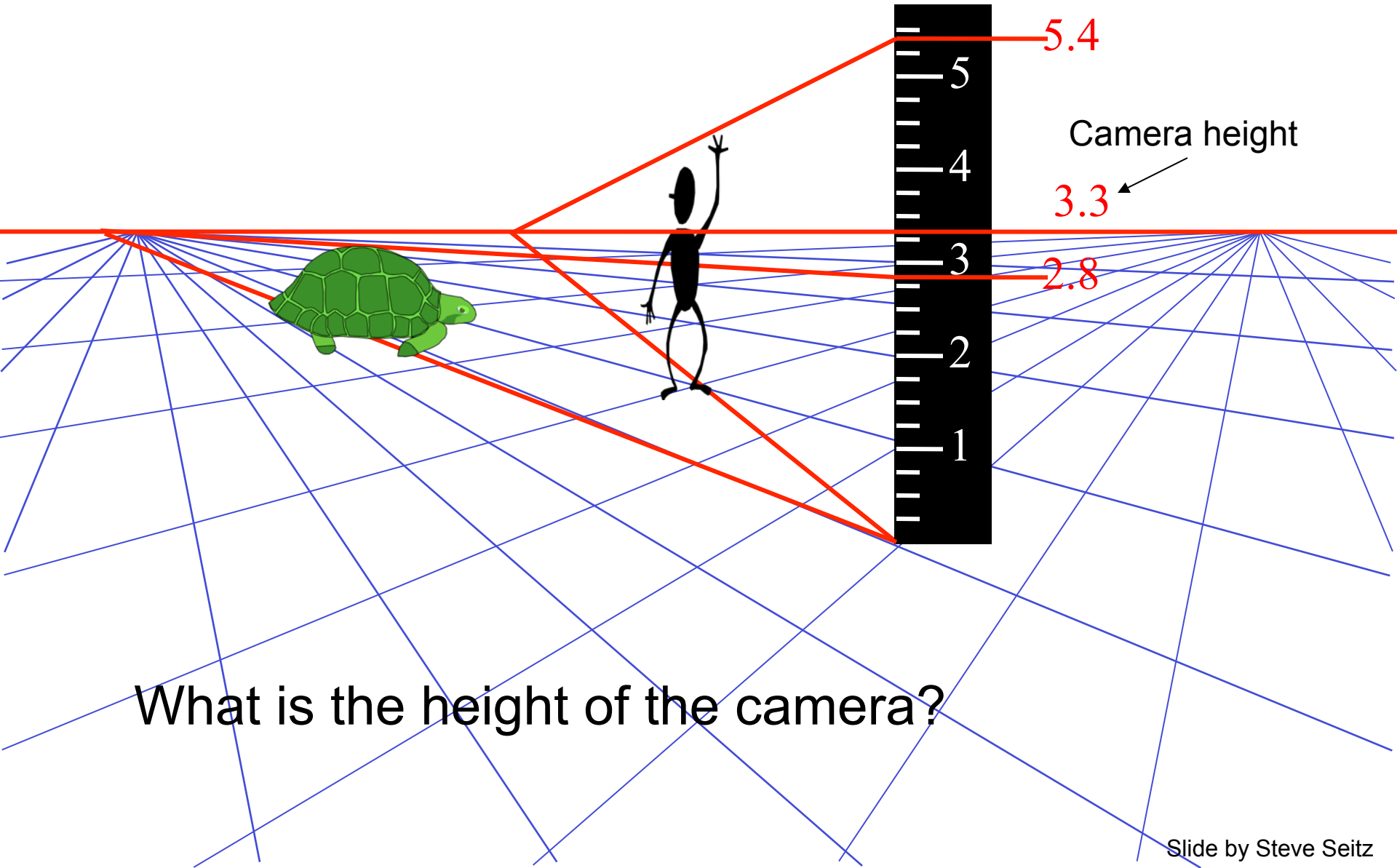


- *Horizon*: vanishing line of the ground plane
 - All points at the same height as the camera project to the horizon
 - Points higher (resp. lower) than the camera project above (resp. below) the horizon
 - Provides way of comparing height of objects

Comparing heights

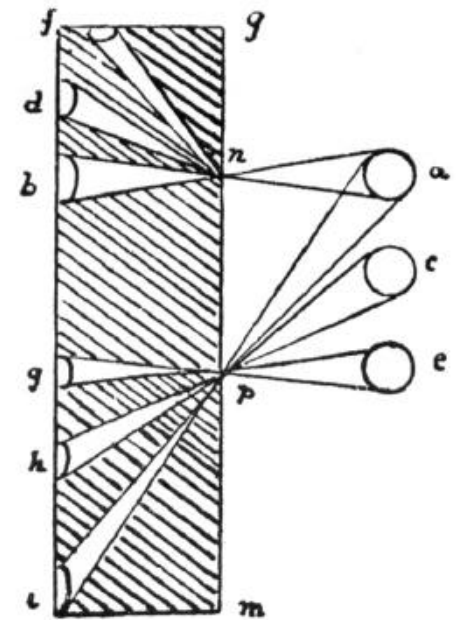
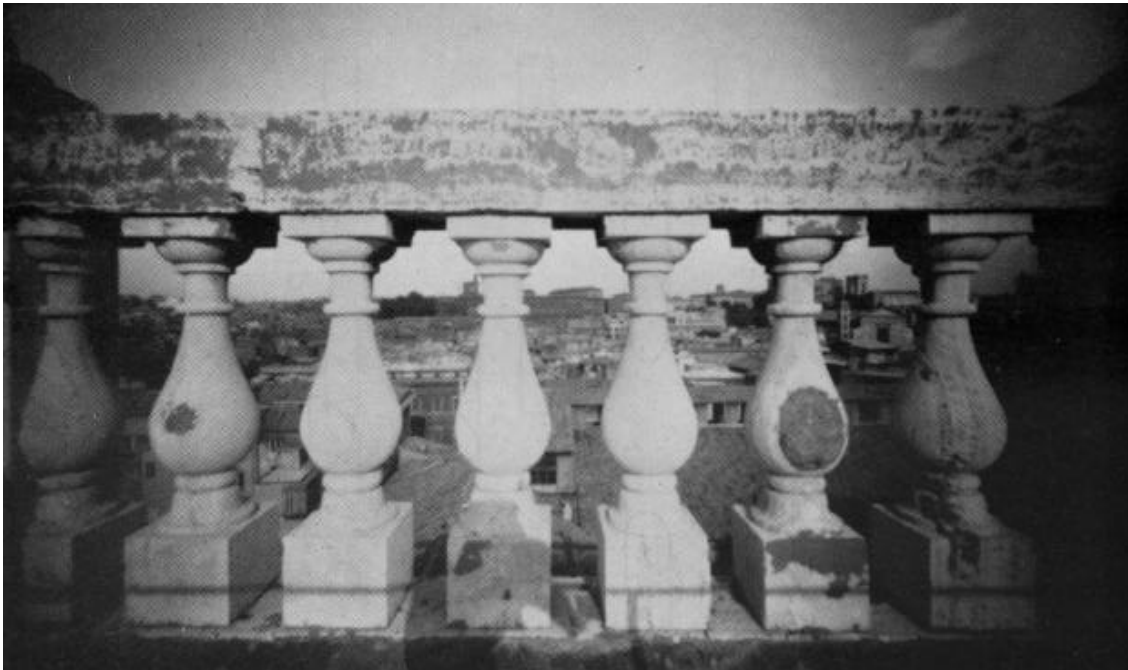


Measuring height



Perspective distortion

- Are the widths of the projected columns equal?
 - The exterior columns are wider
 - This is not an optical illusion, and is not due to lens flaws
 - Phenomenon pointed out by Da Vinci



Perspective distortion

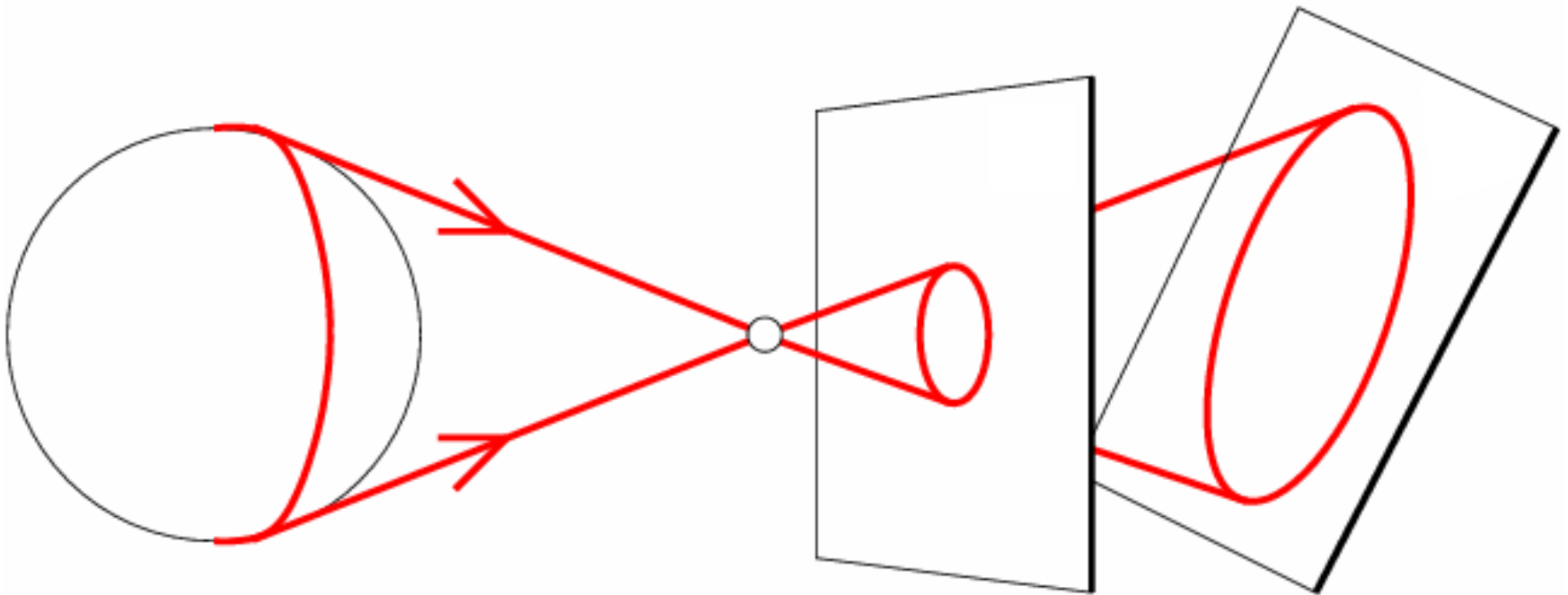
- What is the shape of the projection of a sphere?



Image source: F. Durand

Perspective distortion

- What does a sphere project to?



Perspective distortion: People

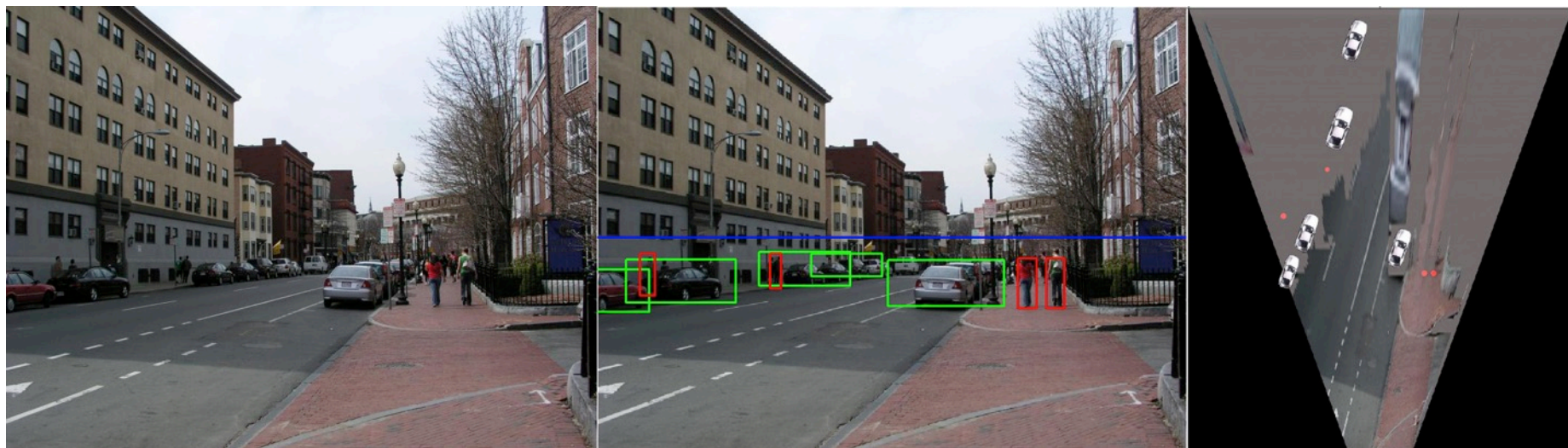


Interlude: when have I used this stuff?

I borrowed these slides from Derek Hoiem, so sometimes “I” means him, and sometimes it means “both of us”

How is this stuff useful?

Object Recognition (CVPR 2006)



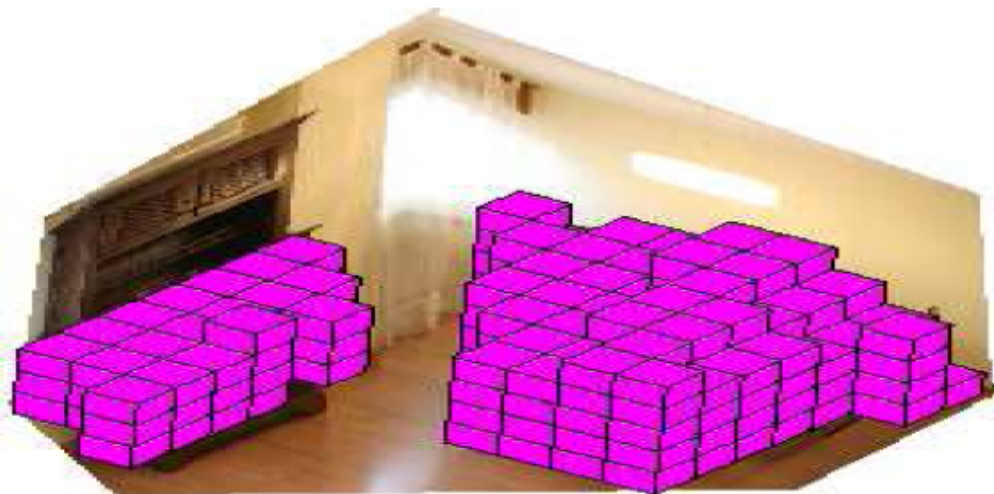
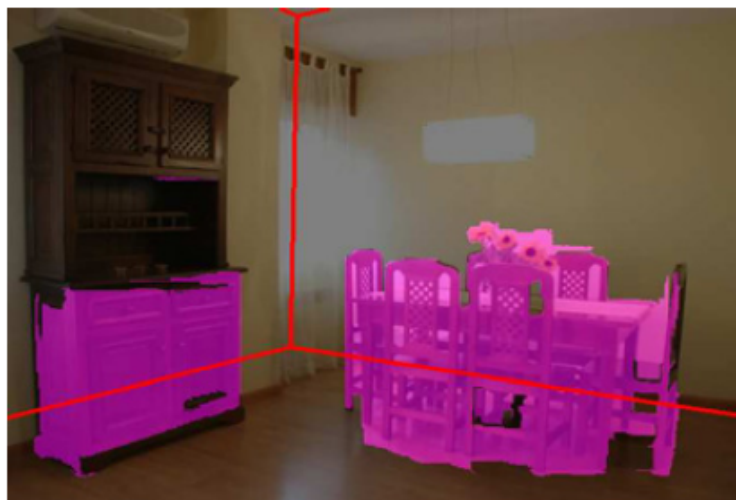
How is this stuff useful?

Single-view reconstruction (SIGGRAPH 2005)



How is this stuff useful?

Getting spatial layout in indoor scenes (ICCV 2009)



How is this stuff useful?

Inserting synthetic objects into images



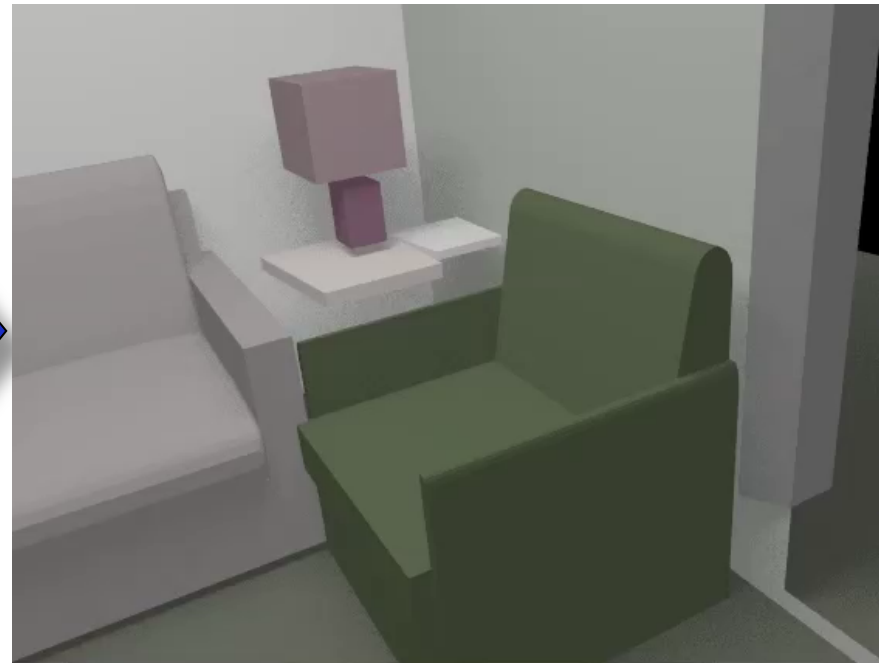
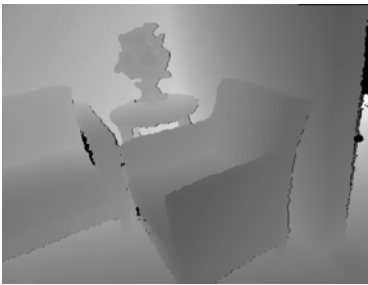
How is this stuff useful?

Inserting synthetic objects into images

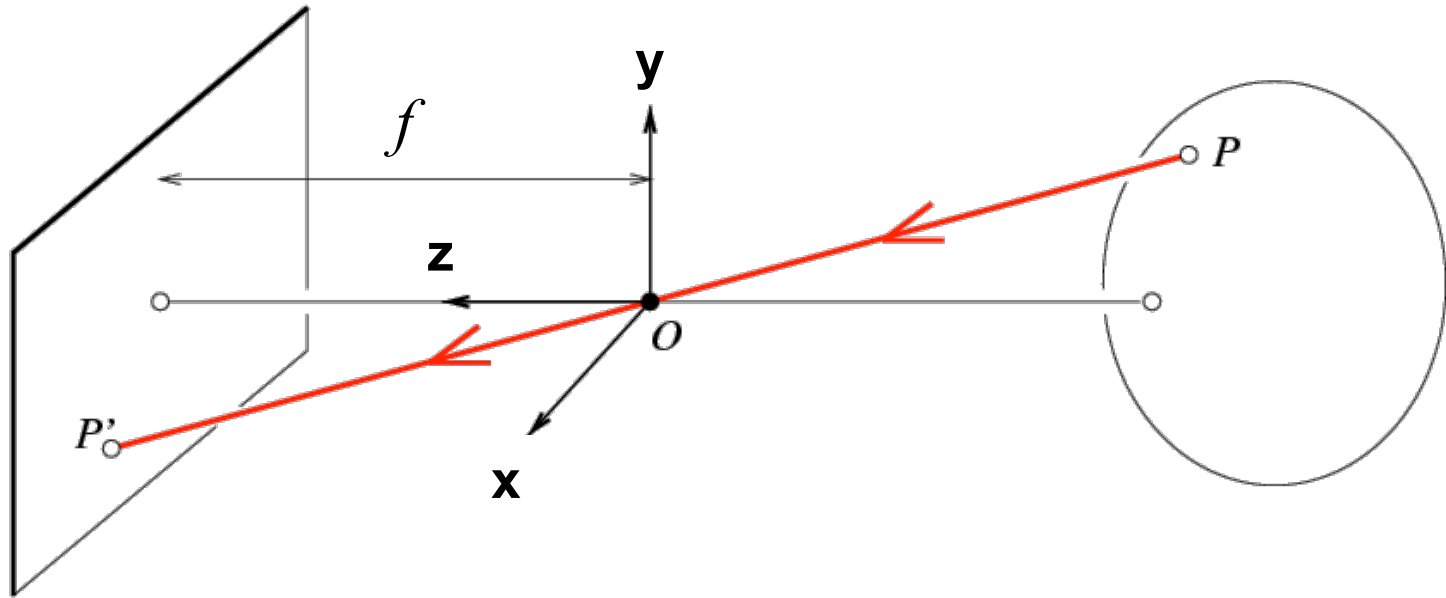


How is this stuff useful?

Creating detailed and complete 3D scene models from a single view (ongoing)



Modeling projection



Projection equation: $(x, y, z) \rightarrow (f \frac{x}{z}, f \frac{y}{z})$

Homogeneous coordinates

$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

Is this a linear transformation?

- no—division by z is nonlinear

Trick: add one more coordinate:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

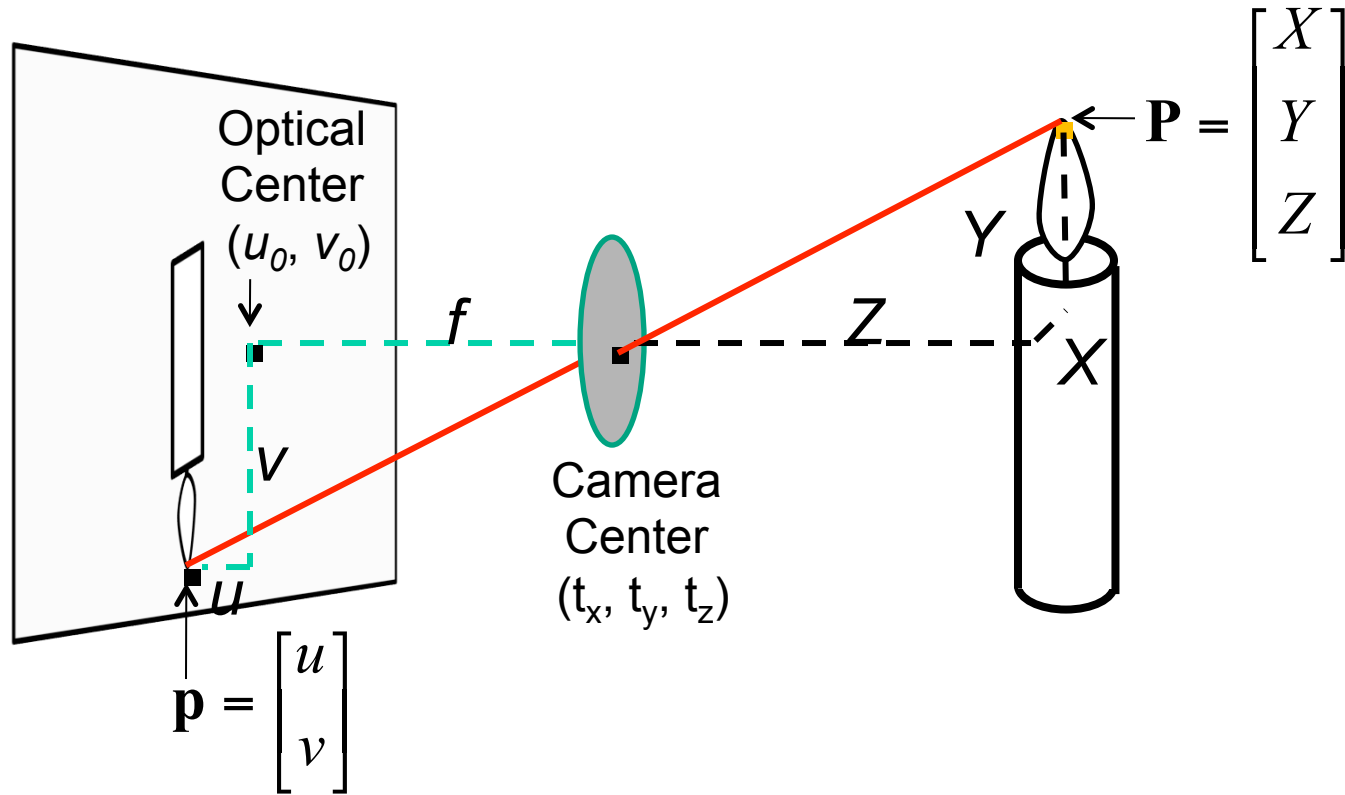
homogeneous scene
coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Projection: world coordinates $\rightarrow$ image coordinates



Homogeneous coordinates

Invariant to scaling

$$k \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kw \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{kx}{kw} \\ \frac{ky}{kw} \\ \frac{kw}{kw} \end{bmatrix} = \begin{bmatrix} \frac{x}{w} \\ \frac{y}{w} \\ 1 \end{bmatrix}$$

Homogeneous Coordinates Cartesian Coordinates

Point in Cartesian is ray in Homogeneous

Basic geometry in homogeneous coordinates

Line equation: $ax + by + c = 0$

$$line_i = \begin{bmatrix} a_i \\ b_i \\ c_i \end{bmatrix}$$

Append 1 to pixel coordinate to get homogeneous coordinate

$$p_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix}$$

Line given by cross product of two points

$$line_{ij} = p_i \times p_j$$

Intersection of two lines given by cross product of the lines

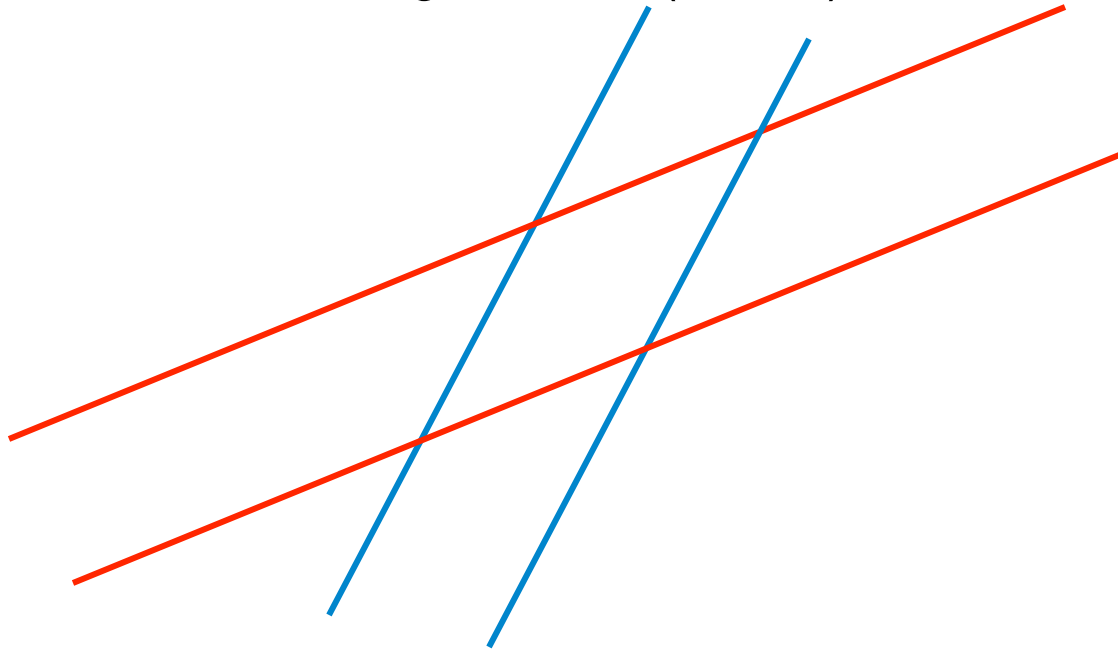
$$q_{ij} = line_i \times line_j$$

Another problem solved by homogeneous coordinates

Intersection of parallel lines

Cartesian: (Inf, Inf)
Homogeneous: $(1, 1, 0)$

Cartesian: (Inf, Inf)
Homogeneous: $(1, 2, 0)$



Perspective Projection Matrix

Projection is a matrix multiplication using homogeneous coordinates

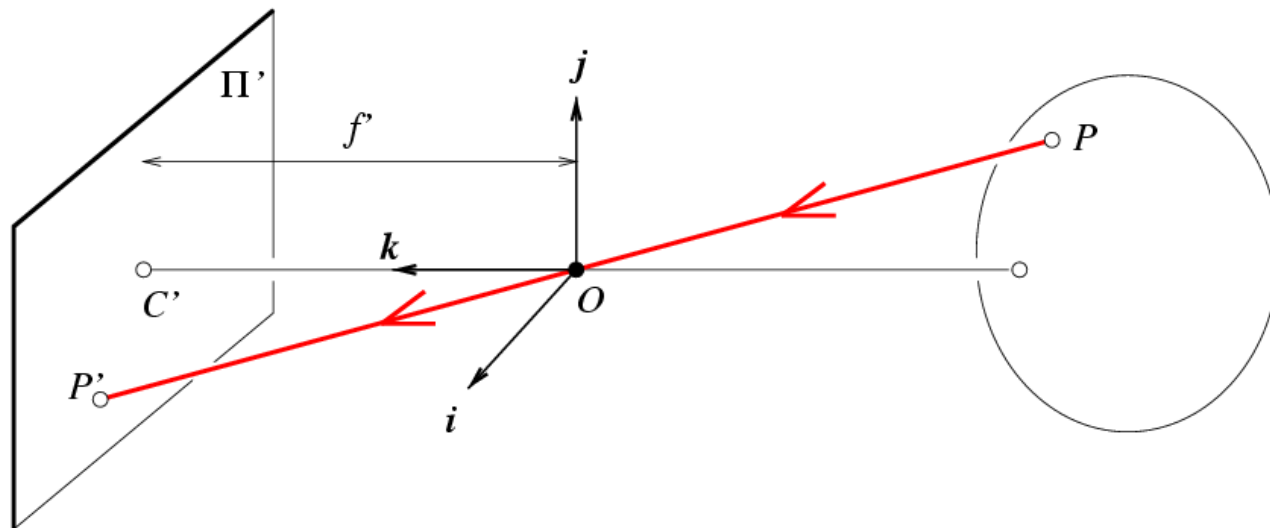
$$\begin{bmatrix} \\ \\ \\ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \Rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

divide by the third coordinate

In practice: lots of coordinate transformations...

$$\begin{pmatrix} \text{2D} \\ \text{point} \\ (3 \times 1) \end{pmatrix} = \begin{pmatrix} \text{Camera to} \\ \text{pixel coord.} \\ \text{trans. matrix} \\ (3 \times 3) \end{pmatrix} \begin{pmatrix} \text{Perspective} \\ \text{projection matrix} \\ (3 \times 4) \end{pmatrix} \begin{pmatrix} \text{World to} \\ \text{camera coord.} \\ \text{trans. matrix} \\ (4 \times 4) \end{pmatrix} \begin{pmatrix} \text{3D} \\ \text{point} \\ (4 \times 1) \end{pmatrix}$$

Projection matrix



Intrinsic Assumptions

- Unit aspect ratio
- Optical center at $(0,0)$
- No skew

Extrinsic Assumptions

- No rotation
- Camera at $(0,0,0)$

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \Rightarrow {}^w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

The matrix $\mathbf{K}$ is the intrinsic camera matrix, and the dashed red box highlights its structure, which corresponds to the intrinsic assumptions listed above.

Remove assumption: known optical center

Intrinsic Assumptions

- Unit aspect ratio
- No skew

Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & u_0 & 0 \\ 0 & f & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Remove assumption: square pixels

Intrinsic Assumptions

- No skew

Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & 0 & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Remove assumption: non-skewed pixels

Intrinsic Assumptions Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & s & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Note: different books use different notation for parameters

Degrees of freedom

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$



$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{matrix} 5 \\ \begin{bmatrix} \alpha & s & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix} \begin{matrix} 6 \\ \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \end{matrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Vanishing Point = Projection from Infinity

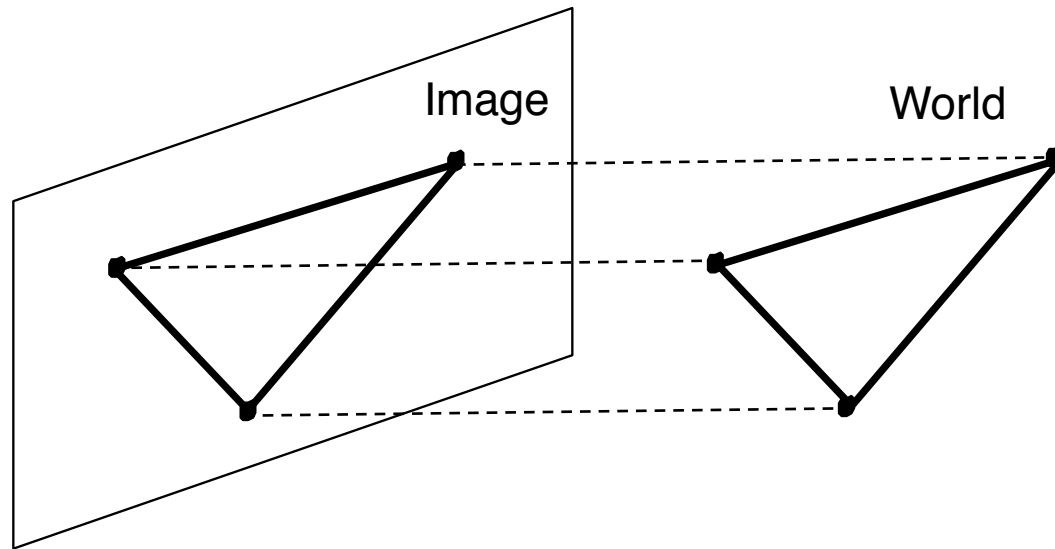
$$\mathbf{p} = \mathbf{K}[\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix} \Rightarrow \mathbf{p} = \mathbf{K}\mathbf{R} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \mathbf{p} = \mathbf{K} \begin{bmatrix} x_R \\ y_R \\ z_R \end{bmatrix}$$

$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_R \\ y_R \\ z_R \end{bmatrix} \Rightarrow \begin{aligned} u &= \frac{fx_R}{z_R} + u_0 \\ v &= \frac{fy_R}{z_R} + v_0 \end{aligned}$$

Orthographic Projection

Special case of perspective projection

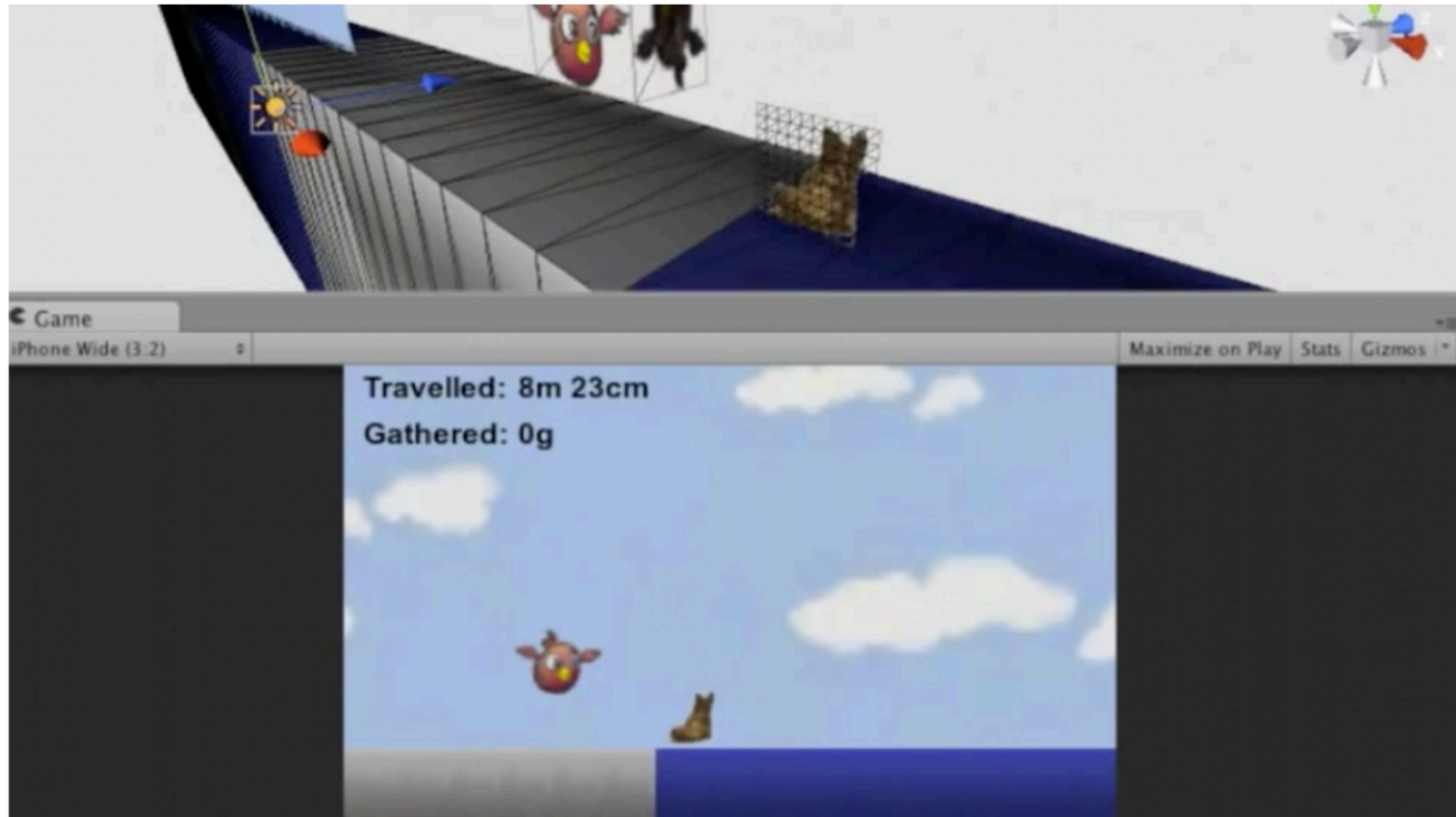
- Distance from center of projection to image plane is infinite
- Also called “parallel projection”



Orthographic Projection

Special case of perspective projection

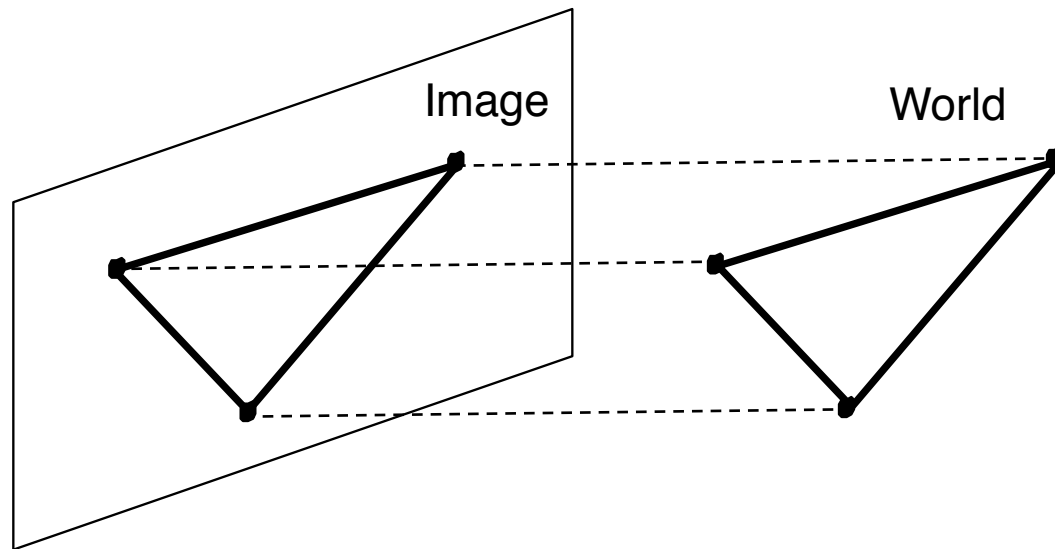
- Distance from center of projection to image plane is infinite
- Also called “parallel projection”



Orthographic Projection

Special case of perspective projection

- Distance from center of projection to image plane is infinite
- Also called “parallel projection”



- What's the projection matrix?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$